

A Resonance-Based Framework for Emergent Matter Formation Conceptual Basis of the Roton Quantum Model

Olav Le Doigt

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Abstract

We introduce the Roton Quantum Model (RQM), a resonance-based conceptual framework aimed at explaining the emergence of matter, interaction, and hierarchy from minimal dynamical principles. Rather than postulating particles, fields, or quantization rules a priori, the model describes physical structure as arising from stable rotational resonance configurations within a continuous energy-density oscillation field.

In this framework, localized and persistent entities emerge through resonance closure, coherence, and inertia-limited reconfiguration. Photons, electrons, nuclei, atoms, and larger-scale structures are interpreted as different organizational levels of the same underlying resonance dynamics, distinguished by internal geometry and rotational inertia rather than by fundamentally different constituents. Conventional forces are understood as effective manifestations of a small set of resonance interaction regimes, including axial entanglement, distance locking, isotropic residual attraction, and energy-density saturation.

The RQM separates global resonance consistency from the finite-speed, inertia-limited motion of physical structures. This distinction provides a constructive interpretation of stability, entanglement, inertia, and hierarchy without invoking point-like entities, singular potentials, or observer-dependent postulates.

The model is presented as a conceptual foundation rather than a closed theory. Its purpose is to identify a coherent ontology and interaction taxonomy from which known qualitative features of physical structure can emerge naturally, and to define a focused program for future analytical and numerical investigation.

1 Introduction and Motivation

The emergence of stable matter from energy remains one of the central conceptual challenges in fundamental physics. While contemporary theories successfully predict experimental outcomes across a wide range of scales, they do so by postulating particles, fields, and interaction laws whose internal structure and origin are not derived, but assumed.

In particular, quantization rules, interaction potentials, and intrinsic properties such as mass, charge, and spin are typically introduced axiomatically or through effective descriptions. As a result, modern physics provides highly accurate answers to the question of *how* physical systems behave, while leaving largely open the question of *why* stable structures exist at all, and why they take the specific forms observed.

This work proposes an alternative conceptual framework in which matter, interaction, and hierarchy are not fundamental ingredients, but emergent consequences of a single organizing principle: rotational resonance in a continuous energy-density field. Rather than beginning with particles or quantized fields, the approach explores how stable, localized, and hierarchically organized structures can arise from resonance closure, coherence, and inertia-limited reconfiguration.

The Roton Quantum Model (RQM) is introduced as a constructive proposal in this direction. Its aim is not to replace the predictive machinery of existing theories, but to provide a deeper explanatory layer beneath them—one in which photons, electrons, nuclei, atoms, and even

large-scale structures appear as different organizational levels of the same underlying resonance dynamics.

The focus of this paper is therefore conceptual rather than phenomenological. We seek to identify minimal assumptions, articulate clear design principles, and demonstrate how a coherent ontology based on resonance can reproduce known qualitative features of physical structure without invoking singularities, point-like entities, or externally imposed quantization rules.

2 Conceptual Gaps and Design Goals

The Roton Quantum Model is motivated not by shortcomings in predictive accuracy, but by persistent conceptual gaps in the foundations of contemporary physical theory. These gaps do not invalidate existing models; rather, they highlight the absence of a simple, constructive picture linking microscopic dynamics, stable structure, and large-scale organization.

2.1 Descriptive success versus constructive explanation

Modern quantum and field-theoretic frameworks describe physical phenomena through abstract state spaces, operators, and effective interaction terms. While extraordinarily successful in practice, these formalisms largely encode observed regularities rather than derive them. Quantization, binding energies, shell structures, and clustering patterns are imposed through boundary conditions or fitted parameters, rather than emerging from an explicit dynamical mechanism in real space.

A central design goal of the RQM is therefore to shift from a descriptive to a constructive perspective: to identify minimal dynamical principles from which stability and structure follow naturally.

2.2 Point-like entities, singular interactions, and renormalization

Many foundational models represent elementary entities as point-like objects interacting through singular potentials. This idealization leads to ultraviolet divergences and the need for renormalization procedures that are mathematically consistent but conceptually opaque. The reliance on infinities and subsequent regularization suggests that essential geometric or structural information is missing from the underlying description.

The RQM is explicitly designed to avoid point-like primitives and singular interaction laws. All entities are assumed to possess finite spatial extent, finite resonance capacity, and finite response times.

2.3 Entanglement, measurement, and observer dependence

In standard quantum mechanics, entanglement is formulated as a property of abstract composite states, with nonlocal correlations postulated rather than derived. The resulting measurement problem introduces a tension between continuous unitary evolution and discontinuous state reduction, often assigning a special role to observation itself.

The RQM adopts a different stance: entanglement is treated as a dynamical consequence of persistent resonance alignment between extended structures. No observer-dependent postulates are required; coherence and decoherence arise from stability or disruption of resonance conditions.

2.4 Spin, inertia, and internal structure

Properties such as spin and inertial mass are introduced in standard theory as intrinsic attributes or representation labels, rather than as consequences of internal dynamics. While this approach is operationally successful, it obscures the physical origin of these quantities and their relation to spatial structure.

A key design goal of the RQM is to interpret inertia and spin geometrically, as manifestations of resistance to reorientation and precession within nested rotational structures.

2.5 Nuclear structure and hierarchy

Nuclear stability, isotope patterns, and the special role of alpha-particle like configurations are well documented empirically, yet are described by a patchwork of phenomenological models. A unifying geometric or dynamical principle operating across nuclear, atomic, and molecular scales remains elusive.

The RQM is constructed to account for hierarchy explicitly: larger structures arise as resonance-stable organizations of smaller ones, without introducing scale-specific laws.

2.6 Design goals of the Roton Quantum Model

Summarizing these considerations, the RQM is guided by the following design principles:

1. all physical entities are extended and finite,
2. interaction arises from resonance rather than force mediation,
3. stability corresponds to resonance closure and energy-density optimization,
4. entanglement is a dynamical alignment phenomenon, not a measurement postulate,
5. inertia limits reconfiguration but not global consistency,
6. the same interaction taxonomy applies across physical scales.

These goals define the conceptual foundation upon which the subsequent sections introduce the specific ontology and dynamics of the Roton framework.

3 Foundational Ontology of the Roton Quantum Model

The Roton Quantum Model (RQM) is built on a minimal ontological framework intended to support the emergence of stable structure, interaction, and inertia without introducing point-like entities, singular potentials, or externally imposed quantization rules. This section defines the fundamental objects and organizing principles of the model. All interaction mechanisms and structural phenomena discussed later are derived exclusively from these foundations.

3.1 The LEDO-Field: Local Energy Density Oscillations

The LEDO-field (Local Energy Density Oscillation field) is introduced as a continuous, spatially extended substrate embedded within spacetime. It is not a force field, carrier medium, or additional dynamical degree of freedom. Rather, it represents the oscillatory background in which local energy-density patterns can form, persist, and interact through resonance.

The LEDO-field supports oscillatory modes characterized by frequency, orientation, and local coherence. Background fluctuations are an intrinsic component of the field and continuously perturb existing configurations. These fluctuations do not act as noise external to the system, but as an inseparable ingredient that enables exploration of nearby resonance-compatible states.

Conventional fields and forces are interpreted as effective, scale-dependent projections of structured resonance behavior within the LEDO-field. No singular sources or point-like excitations are assumed.

3.2 Rotons as Fundamental Resonant Units

A *Roton* is defined as a stable, self-sustaining rotational configuration of energy within the LEDO-field. Kinematically, a Roton is characterized by a finite rotational radius, an intrinsic frequency, an orientation axis, and a finite resonance bandwidth. Its defining property is the existence of a closed rotational trajectory that remains coherent under background perturbations. Dynamic influences may induce a persistently precessing rotation axis in a Roton.

Rotons possess a finite spatial extent and a corresponding *rotonal inertia*, defined as resistance to changes in the orientation or geometry of the rotational trajectory. This inertia governs the temporal response of Rotons to changes in the surrounding resonance environment. Acceleration and reorientation therefore require finite time, even when resonance conditions change.

Rotons act as localized sources of structured contributions to the resonance coupling potential of the LEDO-field. At the same time, they remain dynamically embedded within the same field and are subject to its global and local constraints.

3.3 Resons and Composite Rotational Structures

When multiple Rotons couple through stable resonance configurations, they form composite structures. A *Reson* denotes a closed resonance arrangement that stabilizes specific relative orientations or separations between Rotons. Resons act as elementary binding modes within the model.

More complex assemblies of Rotons linked by one or more Resons form higher-order composite units. Such composites retain a finite spatial extent and collective inertia and may behave as effective single entities at larger scales. Later sections refer to these grid structures as quon- or cage-like configurations, depending on their role in confinement and structural organization.

3.4 Resonance as a Global Consistency Condition

A central distinction of the RQM is the separation between *resonance conditions* and *physical dynamics*. Resonance alignment is not modeled as a propagating interaction or recursive adjustment process. Instead, it is treated as a global consistency condition of the coupled system.

In particular, *entangled resonance alignment* between Rotons is represented as a coherence constraint that may be expressed bi-temporally through phase relationships defined over the system's history. This representation does not imply signal transmission or dynamical influence outside causal limits. It specifies which configurations are resonance-compatible, not how systems move toward them.

Physical motion, reorientation, and structural rearrangement of Rotons occur exclusively through inertia-limited dynamics within the LEDO-field. Finite response times arise from rotonal inertia and background fluctuations. Observable behavior therefore remains causal and time-extended, even though resonance conditions themselves are defined globally.

3.5 Scope and Closure of the Ontology

The LEDO-field, Rotons, and Resons constitute the complete ontological basis of the Roton Quantum Model. No additional particles, forces, or quantization rules are introduced. All interaction regimes described in Sec. ?? and all emergent structures discussed in later sections arise from geometric and dynamical combinations of these elements.

This ontology is intentionally finite and closed. Its purpose is not to replicate existing formalisms at the axiomatic level, but to provide a constructive foundation from which stability, interaction, and hierarchy emerge naturally.

Terminology and scope

Throughout this work, the following terms denote specific levels of rotational resonance organization within the LEDO-field:

- *Roton*: a single-axis rotational degree of freedom representing a localized, self-sustaining rotational resonance with finite spatial extent and rotational inertia.
- *Reson*: an effective resonance coupling, that is, a stable resonance-locked configuration between two or more Rotons, defining a persistent coupling relation.
- *Quon*: a compact assembly of multiple Rotons occupying a common center within their spatial extent, distinguished by the simultaneous realization of multiple resonance channels. A Quon acts as an effective higher-level rotational unit and may be interpreted as a rotating grid node whose structure is defined by the concurrent use of several resonance coupling pathways and is associated with a higher-span resonance coupling potential.

For clarity, several additional terms used throughout this work are defined here in their specific modeling sense.

A *resonance coupling potential* characterizes the span and qualitative nature of a resonance coupling that a Roton offers to its environment. It represents a possible coupling pathway rather than an active interaction. Such a potential may become effective by forming a realized resonance channel, or it may remain unused without producing any dynamical effect.

A *resonance channel* denotes a specific realized pathway of resonance coupling drawn from the available resonance coupling configurations. It refers to a geometrically and phase-compatible alignment between rotational structures through which coherence and stability may be maintained. Resonance channels do not represent particle exchange, energy transport, or classical interaction channels.

Energy-density saturation denotes the local exhaustion of resonance capacity within a finite region of the LEDO-field. In this regime, additional rotational coupling increases local energy-density gradients and produces an effective repulsive response. Rotons of different span contribute to and interact with the local energy-density distribution, and saturation of this distribution limits further compatible coupling.

These terms are introduced as constructive modeling concepts and are not intended to coincide directly with similarly named objects in other theoretical contexts.

4 Resonant Interaction Taxonomy

Within the Roton Quantum Model (RQM), interactions are not introduced as fundamental force laws. Instead, they arise as effective manifestations of resonance constraints and gradients within the LEDO-field. Each Roton contributes a spectrally and directionally resolved resonance potential, and the superposition of these contributions determines the torque and translational tendencies experienced by other Rotons. Which observable interaction emerges depends on the relative orientation, phase relations, and which degrees of freedom are dynamically free to respond.

All interaction phenomena relevant to atomic, nuclear, and large-scale structure can be organized into four principal resonance regimes. These regimes form a complete and finite taxonomy; all subsequent structural emergence described in this work relies exclusively on combinations of these mechanisms.

4.1 Axial Entanglement: Distance-Independent Coupling

The strongest and most constrained interaction arises when two Rotons share a common rotation axis and are locked in an anti-parallel configuration. In this case, the axial resonance channel closes upon itself: resonance contributions cancel toward the exterior while remaining finite internally. The resulting configuration exhibits persistent binding that is effectively independent of spatial separation, as long as axial alignment is maintained.

This regime defines *entangled resonance alignment*. It is modeled as a global consistency condition of the coupled system rather than as a propagating interaction. No signal transmission

or local mediation is implied. Observable motion of the Rotons remains inertia-limited and causal, but the coherence constraint itself does not depend on distance.

Axially entangled Roton pairs behave as composite units with reduced external resonance signature. Chains or clusters of such entanglements require spatial overlap within the relevant rotational radii and form the basis of confined, nuclear-scale structures discussed in later sections.

4.2 Parallel-Axial Resonance and Distance Locking

If two Rotons share a common or nearly common axis without anti-parallel entanglement, their resonance potentials interfere along that axis. Constructive and destructive interference select discrete equilibrium separations at which the net resonance torque vanishes. These separations define preferred distances

$$d \approx n d_0(R, f, \Delta\phi), \quad n \in \mathbb{Z},$$

where d_0 depends on the rotational radius R , intrinsic frequency f , and relative phase $\Delta\phi$.

This regime produces *distance-locking interactions*. Unlike axial entanglement, the coupling strength depends on separation and alternates between attractive and repulsive domains. The interaction is effectively one-dimensional along the shared axis and supports linear chains, layered grids, and shell-like arrangements.

Depending on constraints, different physical responses dominate: free Rotons adjust both phase and position, position-locked Rotons adapt phase, and phase-locked Rotons respond primarily through displacement. All cases arise from the same underlying resonance structure.

4.3 Isotropic Residual Resonance Attraction

When rotational axes fluctuate rapidly or remain uncorrelated, directional resonance contributions average out. The remaining interaction is approximately isotropic and decays with distance. This regime represents the coarse-grained remnant of transient and non-coherent alignments.

The resulting attraction is weak but long-ranged and does not rely on persistent axial structure. It provides a universal background coupling between Rotons and dominates only where more structured resonance channels are absent. In later sections, this contribution is associated with effective large-scale attractive behavior once higher-order resonance organization is averaged out.

4.4 Energy-Density Saturation and Short-Range Repulsion

At sufficiently small separations, the local resonance capacity of the LEDO-field becomes saturated. Additional Rotons introduced into such a region no longer find attractive resonance channels. Instead, they increase local energy-density gradients, producing an effective repulsive response.

This short-range repulsion enforces finite minimal distances between Rotons and prevents singular collapse. Its strength rises rapidly as separation decreases relative to the characteristic rotational radius. Because all Rotons possess finite spatial extent, no divergence occurs. This mechanism replaces hard-core exclusions or singular potentials by a smooth, physically interpretable saturation effect.

4.5 Summary of the Interaction Hierarchy

The effective interaction between Rotons can be summarized schematically as

$$F = F_{\text{ent}} + F_{\parallel} + F_{\text{iso}} + F_{\text{rep}},$$

where axial entanglement dominates when present, parallel-axial resonance governs distance locking, isotropic residual attraction provides a weak long-range background, and energy-density saturation enforces short-range exclusion.

No additional interaction types are required. Atomic, nuclear, molecular, and large-scale structures arise from different geometric and dynamical combinations of these four resonance regimes. All couplings remain finite, size-dependent, and free of singular behavior.

5 Hierarchical Structural Emergence

Having defined the foundational ontology (Sec. ??) and the complete interaction taxonomy (Sec. ??), we now describe how stable physical structures emerge as hierarchical resonance configurations. No additional mechanisms are introduced. All structures arise from geometric and dynamical combinations of Rotons coupled through the resonance regimes already defined.

5.1 Bootstrapping Photon-Like Structures

The simplest stable configuration in the Roton framework corresponds to a single-axis Roton whose rotational trajectory remains coherent under background fluctuations. Such a structure exhibits minimal rotonal inertia and propagates freely as long as its rotation axis remains stable. It may be interpreted as a soliton-like, self-sustaining pattern embedded in the underlying LEDO-field.

Photon-like behavior is identified with this class of single-axis resonance-closed configurations, characterized by coherence, low inertia, and unrestricted propagation along a stable axis.

5.2 Electron-Like Roton Structures

The first level of multi-rotational organization arises from interlocked rotations formed by entangled photon-like Rotons. Such configurations are initially near-planar, but already extend into an additional dimension through the rotation of the underlying constituent Rotons.

Electron-like entities emerge as multi-axis composite structures in which several Rotons are mutually coupled while sharing a common center within their rotonal span. In the minimal stable case, three rotational degrees of freedom are locked into a self-sustaining configuration. This tri-axial structure exhibits substantially higher rotonal inertia, since changes in motion require coordinated reorientation of multiple rotation axes.

Charge-like interaction behavior arises from the structured resonance coupling availability associated by this configuration. The distinction between photon-like and electron-like entities is therefore not attributed to different fundamental constituents, but to different levels of internal resonance organization.

5.3 Quon Grid Structures

At higher resonance densities, electron-like Rotons may assemble into compact composite units that offer multiple potential resonance channels into different spatial directions. Such assemblies act as nodes within higher-level, grid-like resonance structures in space.

When such a node itself undergoes coherent rotation, it constitutes a higher-level Roton, referred to as a *Quon*. Quons may participate in further resonant organization, initially forming entangled Quon pairs and subsequently extended grid structures through shared resonance channels.

5.4 Nuclear-Scale Resonant Cages

Certain compound Quon arrangements act as resonance cages, in which internal rotational degrees of freedom are stabilized and constrained by the surrounding resonance geometry. Proton-like configurations are interpreted as such cages, where an electron-like Roton structure is confined in orientation and phase. The resulting external resonance pattern constitutes an effective coupling channel for orbital electrons without exposing free electron-like behavior.

Neutron-like configurations correspond to alternative internal balances of the same resonance elements, differing primarily in how external coupling channels are expressed. No fundamentally distinct constituents are required.

Highly symmetric paired Quon assemblies form exceptionally stable cores. Proton-, neutron-, and alpha-particle-like configurations maximize compatible resonance channels while minimizing local energy-density saturation and repulsive gradients. Their robustness makes them natural building blocks for larger nuclear structures and provides a constructive explanation for why such configurations naturally arise in the formation of atomic structures.

5.5 Atomic Orbitals as Resonance-Locked Trajectories

Atomic structure emerges when electron-like Rotons couple to nuclear-scale resonance cages. Discrete orbital radii arise as resonance-compatible trajectories at which the integrated torque on the electron's rotational axes vanishes on average. These radii act as attractors of the relaxation dynamics under background fluctuations and correspond to shell structure.

Within a given shell, multiple electrons coexist by occupying distinct rotational planes and phase relations. When two electrons share a compatible radius, they may form paired configurations by arranging their internal phases such that external resonance signatures are partially suppressed while internal coherence is enhanced.

Beyond planar motion, redistribution of rotational activity across axes allows for closed three-dimensional trajectories. Shell differentiation therefore reflects differences in resonance geometry rather than imposed angular-momentum quantum numbers.

5.6 Resonance Induction and Molecular Bonding

Electron-like Rotons can simultaneously participate in multiple resonance channels, corresponding to their internal rotational degrees of freedom. In molecular environments, valence electrons establish shared resonance paths that couple multiple nuclear cores into a single coherent structure.

This process is referred to as *resonance induction*. Existing stable resonance configurations attract additional compatible degrees of freedom and reorganize them into larger, energetically favorable arrangements. Bond lengths and angles are determined by the requirement that shared electron trajectories remain phase-coherent and torque-balanced with respect to all participating centers, while avoiding local resonance saturation.

Chemical regularities thus emerge as geometric constraints of multi-center resonance closure rather than as consequences of predefined force laws.

5.7 Large-Scale Structures and Effective Forces

At still larger scales, the same interaction taxonomy applies in coarse-grained form. When detailed axial alignments and distance-locking structures average out, isotropic residual resonance attraction dominates. This regime yields effective long-range attractive behavior between composite structures.

Rotational coherence on galactic scales contributes additional stabilization through collective resonance effects, favoring filamentary and disk-like arrangements. From this perspective, part of the dynamics commonly attributed to dark components may be interpreted as emergent behavior of large-scale rotational resonance organization within the LEDO-field, without requiring new particulate matter.

5.8 Hierarchy and Closure

Across all scales, from photon-like Rotons to galactic structures, stability arises from the same four resonance regimes defined in Sec. ?? . Higher-level entities differ only in internal organization,

collective inertia, and the degree to which resonance structure is exposed or internally confined.

No new interaction principles appear at higher complexity. Hierarchy reflects resonance geometry, not a change of physical law.

6 Numerical Indicators and Model Validation

The present formulation of the Roton Quantum Model is primarily conceptual, yet it admits clear criteria for internal validation and falsification through numerical experimentation. Because all interaction mechanisms are geometric and resonance-based, the model lends itself naturally to direct simulation of Roton ensembles under controlled initial conditions.

Existing proof-of-principle simulations consistently exhibit the following qualitative features:

- spontaneous emergence of discrete equilibrium distances rather than continuous separations,
- formation of stable chains, grids, and shell-like arrangements from initially unstructured configurations,
- persistence of axial entanglement under spatial separation,
- absence of singular collapse due to short-range energy-density saturation,
- exceptional robustness of symmetric quon-compounds creating alpha-particle-like cores,
- inertia-limited relaxation dynamics following changes in resonance constraints.

These behaviors arise without externally imposed quantization rules, predefined potentials, or hard exclusion constraints. They follow directly from the finite span of Rotons and the resonance regimes defined in Sec. ??.

For quantitative characterization, several numerical indicators are particularly relevant: distance spectra of Roton pairs, lifetime distributions of composite structures, axis-alignment order parameters, local energy-density statistics, and response times to controlled perturbations. Together, these observables provide a bridge between roton-level dynamics and experimentally accessible atomic, nuclear, or mesoscopic phenomena.

Long-duration simulations starting from quasi-homogeneous conditions represent a critical next step. Of particular interest is whether electron-like, nucleon-like, and atomic-scale structures emerge as attractors of the dynamics without scale-specific tuning.

7 Open Research Directions

The present formulation of the Roton Quantum Model is intentionally compact at the axiomatic level, yet structurally rich in its emergent consequences. Several concrete research directions arise naturally from the framework and delineate clear paths toward quantitative testing, refinement, and possible falsification.

Analytical formulation of resonance coupling

A primary task is the derivation of closed analytical expressions for resonance-induced interaction terms within the LEDO-field. This includes explicit functional forms for co-axial, parallel-axial, isotropic, and saturation-driven contributions, as well as their dependence on rotational radius, frequency, bandwidth, and phase relations. A tensorial representation of resonance coupling would enable direct comparison with established field-theoretic descriptions in appropriate limits.

Scaling relations across physical regimes

The model predicts that atomic, nuclear, and large-scale structures arise from the same interaction taxonomy, differentiated only by scale and rotational inertia. Systematic derivation of scaling relations connecting microscopic Roton parameters to effective macroscopic observables (orbital radii, binding energies, relaxation times) is therefore essential for quantitative validation.

Long-duration numerical simulations

Existing simulations demonstrate local stability, distance locking, and shell formation. A key next step is long-duration, large-ensemble simulations starting from quasi-homogeneous initial conditions, designed to test whether electron-like, nucleon-like, and atomic structures emerge spontaneously without scale-specific tuning. Particular emphasis should be placed on robustness under perturbation and sensitivity to background fluctuations.

Resonance stability and isotope structure

Within the RQM, nuclear stability corresponds to the existence of closed resonance networks among compact Roton assemblies. Systematic exploration of these configurations may clarify why certain isotope combinations are favored while others are unstable, and whether known patterns (such as alpha clustering and magic numbers) correspond to resonance-complete geometries.

Orbital dynamics and molecular structure

Further work is required to map resonance-locked electron trajectories onto observed atomic and molecular geometries. This includes identifying how multi-axis precession, pairing, and resonance induction constrain bond lengths, angles, and coordination numbers across chemical systems, without invoking externally imposed orbital bases.

Connections to established formalisms

Finally, it is important to establish systematic connections between the resonance-based language of the RQM and conventional quantum and field-theoretic formalisms. This includes deriving effective Hamiltonians, identifying approximate Hilbert-space representations of stable Roton configurations, and clarifying which aspects of standard theory emerge as coarse-grained limits of the underlying resonance dynamics.

Together, these directions define a focused research program aimed at transforming the Roton Quantum Model from a coherent conceptual framework into a quantitatively testable theory.

8 Conclusions

This work has introduced the Roton Quantum Model as a resonance-based, deterministic framework for the emergence of matter, interaction, and hierarchy across physical scales. Rather than postulating particles, forces, or quantization rules a priori, the model derives stability and structure from the geometry of rotational resonance within a continuous energy-density oscillation field.

By separating global resonance constraints from inertia-limited local dynamics, the framework provides a constructive interpretation of phenomena commonly treated as axiomatic in conventional theories. Photons, electrons, nuclei, atoms, molecules, and large-scale structures are understood as different organizational levels of the same underlying Roton ontology, distinguished by internal resonance geometry and collective inertia rather than by fundamentally distinct constituents.

All interaction behavior is shown to arise from a finite taxonomy of resonance regimes: axial entanglement, parallel-axial distance locking, isotropic residual attraction, and energy-density saturation. No singularities, point-like entities, or externally imposed potentials are required. Hierarchy reflects resonance organization, not a change of physical law.

The Roton Quantum Model is not presented as a closed or final theory. Its value lies in offering a coherent, testable alternative perspective that emphasizes constructive mechanisms over descriptive formalisms. Future work must establish quantitative scaling relations, perform large-scale simulations of spontaneous structure formation, and identify clear experimental signatures capable of confirming or falsifying the framework.

In this sense, the model proposes a shift in viewpoint: from a universe assembled from elementary objects governed by predefined interactions, to one in which structure, inertia, and interaction emerge naturally from resonance, rotation, and coherence as fundamental organizing principles.

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Author Information

Author: Olav Le Doigt

Profession: Dipl. El. Ing. ETH

ORCID: [0009-0002-3356-9634]

Contact: [olav@ledoigt.ch]

Web: ledoigt.ch

The author is an independent researcher with a long-standing interest in foundational aspects of quantum theory, field dynamics and emergent phenomena in complex systems.

Companion Website and Further Material

A non-technical introduction, extended explanations, graphical illustrations and ongoing discussions of the Roton Quantum Model are available on the project website:

- Roton Quantum Model – intuitive overview and guided tour: <https://ledoigt.ch>
- Technical notes, simulations and updates: <https://ledoigt.ch/ledoigt/notes>
- Discussion and feedback channel: <https://ledoigt.ch/ledoigt/discuss>

These resources are intended as a more accessible companion to the present article, offering step-by-step motivation, visualizations, and evolving refinements of the model.

Data and Code Availability

Simulation scripts, illustrative numerical experiments and example configurations of Rotons used to explore the qualitative behavior reported in this work are (or will be) made available at:

- Code repository (numerical experiments): <https://github.com/ledoigt>
- Supplementary animations and figures: <https://ledoigt.ch/ledoigt/collections/>

Interested readers are invited to reproduce, extend and critically test the presented ideas.

Author's Note

This work is intended as an open invitation to the theoretical physics and complex-systems communities to examine, challenge and refine the Roton Quantum Model. The author explicitly welcomes critical assessments, alternative formulations and concrete proposals for experimental or numerical tests, which may help to clarify the model's scope, limits and possible connections to established frameworks.